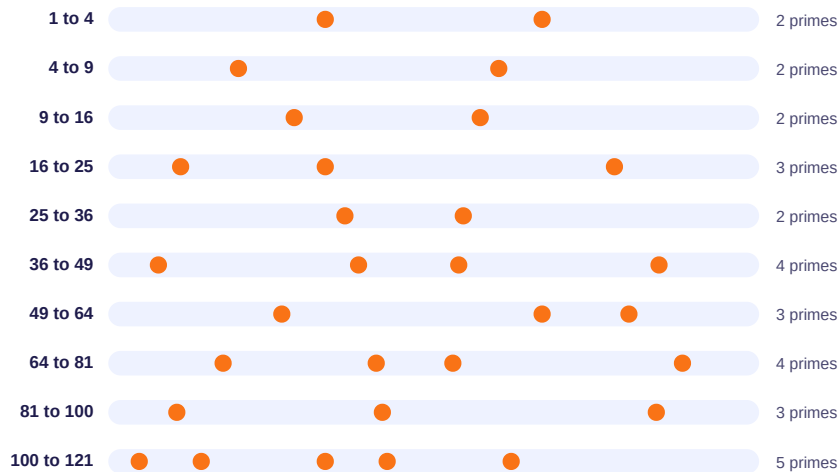


Legendre's Conjecture

Between any two perfect squares that sit next to each other, there's always a prime.



Each strip runs from one square to the next. Orange dots are primes. Every strip has at least two so far.

TRY IT

1 to 4: 2, 3

4 to 9: 5, 7

9 to 16: 11, 13

16 to 25: 17, 19, 23

100 to 121: 101, 103, 107, 109, 113

WHAT MATHEMATICIANS KNOW

- It is named for the French mathematician Adrien-Marie Legendre. In 1912 Edmund Landau listed it, with Goldbach and twin primes, among four prime problems that seemed out of reach. All four are still open.
- A related fact is proved: there is always a prime between any number and its double (Pafnuty Chebyshev, around 1850).
- There is always a prime between two consecutive cubes, but this has only been proved for very large numbers.

WHY IT'S STILL UNSOLVED

The gap from one square to the next keeps growing, which should make primes easier to find there. But primes also have long gaps between them, and no one has proved the prime gaps never outgrow the gaps between squares.

TRY THIS WITH STUDENTS

- Have students draw the strips on graph paper, one per row, and count the primes in each.
- Ask: How long is the strip that starts at 5×5 ? At 6×6 ? Can you predict any strip's length? (It's always one more than twice the number being squared.)
- Ask: Which strip has the fewest primes so far? Could a strip ever have none?

WORDS TO KNOW

Perfect square a whole number times itself, like 1, 4, 9, 16

Consecutive one right after another